Cisco PE Node MAR-PE-13 Standard Operating Procedure for Minor Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding)

Node: MAR-PE-13 | **Vendor:** Cisco | **Severity:** Minor | **Alarm Code:** ALARM_2905

Description

The Marathahalli Core PE node MAR-PE-13 is a Cisco ASR 9000 series router, running Cisco IOS XR Software. The PE node is configured as a border router, participating in multiple routing protocols including BGP, OSPF, and ISIS. The failure domain of routing protocol flap on this node is a common issue that can occur due to various reasons such as misconfiguration, hardware failure, or software bugs. The BGP route churn is caused by the periodic refresh of BGP routes, which can lead to increased CPU utilization and memory usage. OSPF adjacency loss can occur due to issues with the OSPF adjacency establishment or maintenance. ISIS LSP flooding is caused by the periodic refresh of ISIS link-state advertisements. The ALARM_2905 is a minor alarm indicating a BGP neighbor flapping, which can be caused by the above-mentioned issues. This SOP outlines the diagnostic and resolution steps to identify and fix the root cause of the routing protocol flap on MAR-PE-13.

Pre-requisite Checks

- Check: Verify that the redundant link to the next-hop router is up and operational.

- Check: Check the Cisco IOS XR Software version running on MAR-PE-13 and ensure it is up-to-date.

- Check: Verify that the BGP, OSPF, and ISIS configurations on MAR-PE-13 are correct and consistent with the rest of the network.

Potential Root Causes

Cause	Probability
BGP neighbor flapping due to misconfiguration	High
Hardware failure causing routing protocol instability	Medium
Software bug causing routing protocol instability	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp Summary, Expected Output: BGP table version is 0, main routing instance, Description: This command displays a summary of the BGP table version, main routing instance, and other relevant information. It helps to identify any issues with the BGP neighbor flapping.
```

```
Command: show ospf neighbor, Expected Output: Neighbor ID Pri State
Dead Time Address Interface, Description: This command displays a list
of OSPF neighbors, their priority, state, dead time, address, and
interface. It helps to identify any issues with the OSPF adjacency
loss.

Command: show isis database, Expected Output: ISIS LSP Database
(Sequence/LSID/AGE), Description: This command displays the ISIS LSP
database, including the sequence number, LSID, and age of the LSP. It
helps to identify any issues with the ISIS LSP flooding.
```

Resolution Steps

Step 1: Step: Verify BGP neighbor configuration, Command: show bgp neighbor , Description: Verify that the BGP neighbor configuration is correct and consistent with the rest of the network.

Step 2: Step: Check for hardware issues, Command: show tech-support, Description: Check for any hardware issues that may be causing the routing protocol instability.

Step 3: Step: Update Cisco IOS XR Software, Command: install system software, Description: Update the Cisco IOS XR Software to the latest version to fix any software bugs that may be causing the routing protocol instability.

Post-Check Verification

```
Command: show bgp Summary, Expected Output: BGP table version is 0,
main routing instance, Description: Verify that the BGP table version,
main routing instance, and other relevant information are correct and
consistent with the rest of the network.

Command: show ospf neighbor, Expected Output: Neighbor ID Pri State
Dead Time Address Interface, Description: Verify that the OSPF
neighbors, their priority, state, dead time, address, and interface
are correct and consistent with the rest of the network.

Command: show isis database, Expected Output: ISIS LSP Database
(Sequence/LSID/AGE), Description: Verify that the ISIS LSP database,
including the sequence number, LSID, and age of the LSP, are correct
and consistent with the rest of the network.
```

Roll-back Procedures

- Step: Restore previous configuration, Command: copy run start, Description: Restore the previous configuration to the running configuration using the copy run start command.

- Step: Revert to previous software version, Command: install system software , Description: Revert to the previous software version using the install system software command with the previous version number.